Project 2: Data is Beautiful

Overview

Data visualisation and representation have creative elements into it and provide a powerful way of communication from raw data and numbers. The project provides an opportunity to innovate in representing data after collecting and processing the data.

Learning Outcomes

- *Introduction to Data Cycle.* The project leads to implementing and experiencing the entire data cycle - collection, processing, and representation.
- *Data Processing. The collected raw data needs to be processed to get it in some tabular form.*
- *Creative Thinking.* Data representation provides openness in designing the graphs and charts.
- *Data Analysis.* Interpreting the charts and graphs to make conclusions.

Material Required

- Sticky notes or Paper pieces
- Small board to pin the paper pieces (notice board)
- Coloured pencils/pens/crayons
- Projector to showcase images (or take colour printouts of data representations given at the end)

Project Plan

Project Activity	Time	Description
Launch	10-15 mins	The teacher will show examples of creative ways of representing diagrams. The students will also share what they understand from the diagram shown.
Explore the Data Cycle	20-25 mins	Classroom activity to record data and process it. The Teacher will demonstrate ways to creatively represent it.
Data is Beautiful	1 week	The students will create innovative ways of representing data. The class will derive conclusions from the graphs/charts.

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Launch

The teacher will show the following images to the class and discuss what they could interpret. The teacher also discussed the impact of creative and effective data representation in communication

1. ***1978 Study of Hand Washing.*** The image representation becomes highly relevant during the pandemic and in the post-pandemic years. It uses the subject of the data, hands, as the location of the data. The study shows the areas most often missed while washing hands.

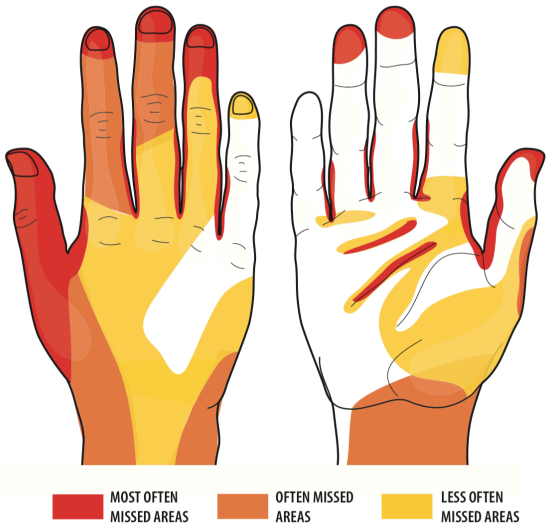


Fig: 1978 Study of Hand Washing
(Source: https://flowingdata.com/2020/02/18/most-often-missed-areas-while-washing-hands/)

2. ***Earth Temperature over Time.*** Play the video (link: https://svsgsfc.nasa.gov/5191/) in the classroom about changes in Earth's temperature over time. NASA represented the data of Earth's temperature over the years with an animation, highlighting concerning changes. Such state-of-the-art visualisation demonstrates how creative visuals can be and how even small changes in pattern, shape, colour, or shading can clearly and crisply communicate the findings.

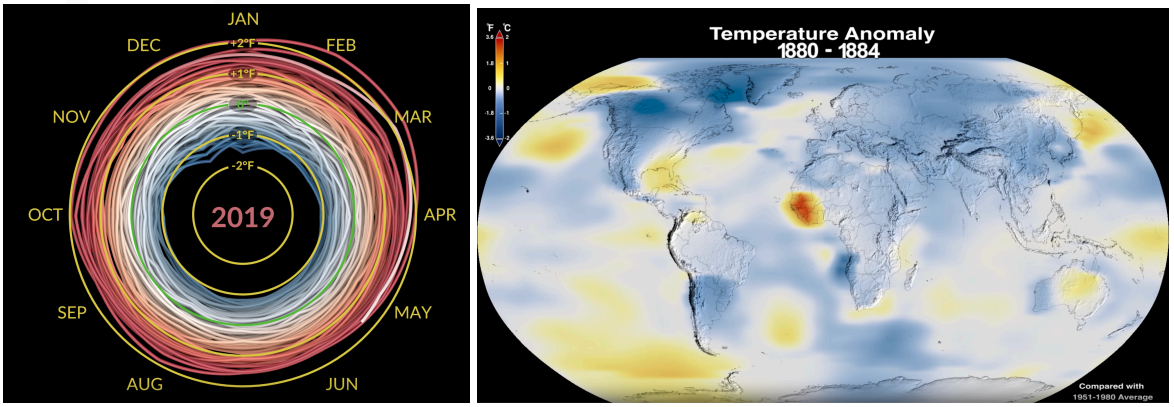


Figure: Earth Temperature over Time
(Source: Scientific Visualization Studio, NASA)

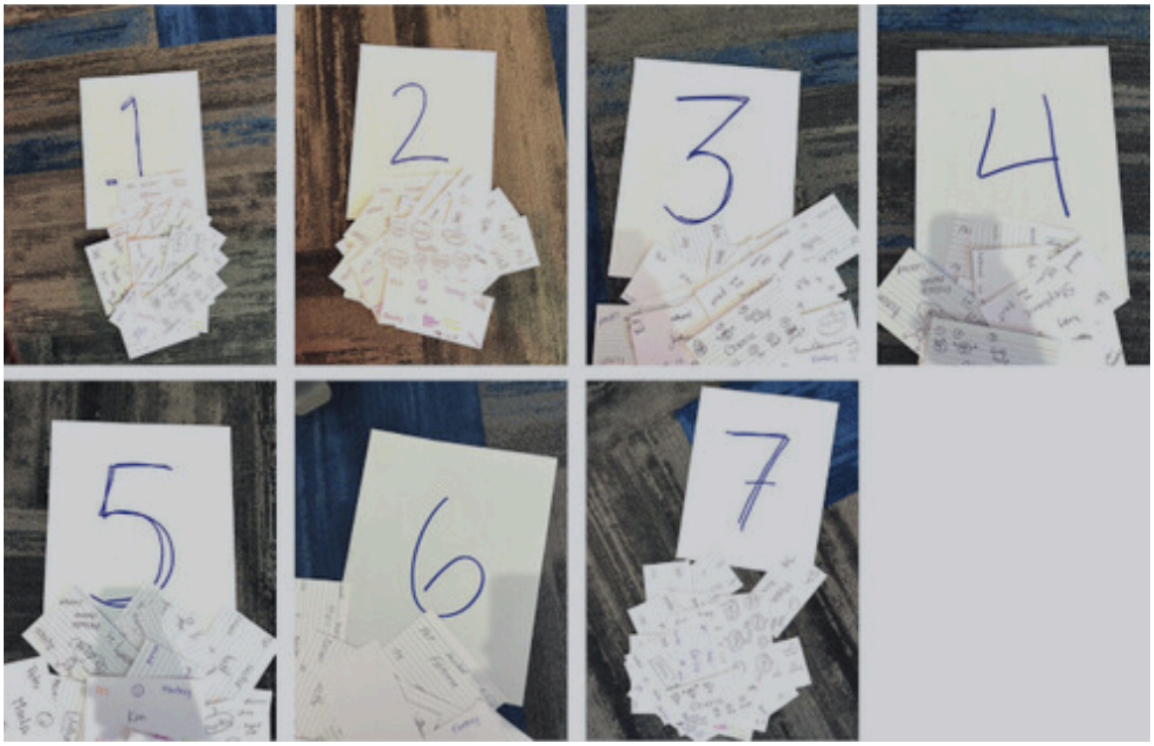
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Explore the Data Cycle

The teacher shall pose the following questions to the students in the classroom. Most question sets focus on one's interests and hobbies.

- ***Data Collection & Processing*.** The students shall answer each in an individual paper with a question label, i.e. 1, 2, etc. The teachers shall collect the answers and stick/pin them on the common board, grouping them in order. After everyone has responded, the students will create data tables for each question.

- ***Data Representation*.** The Teacher will show the following diagrams as an example to students. Let the students come up with different creative ways to represent the data collected in the classrooms.



Possible set of questions:

1. What makes you happy at home?
2. What emotion are you feeling today before coming to school?
3. What type of books do you read the most?
4. What is the strongest emotion you feel when doing art in the classroom?
5. How many sit-ups can you do in 1 minute?
6. How much cycling can you do in a stretch?
7. Who is someone you admire?
8. Which sports do you enjoy watching?

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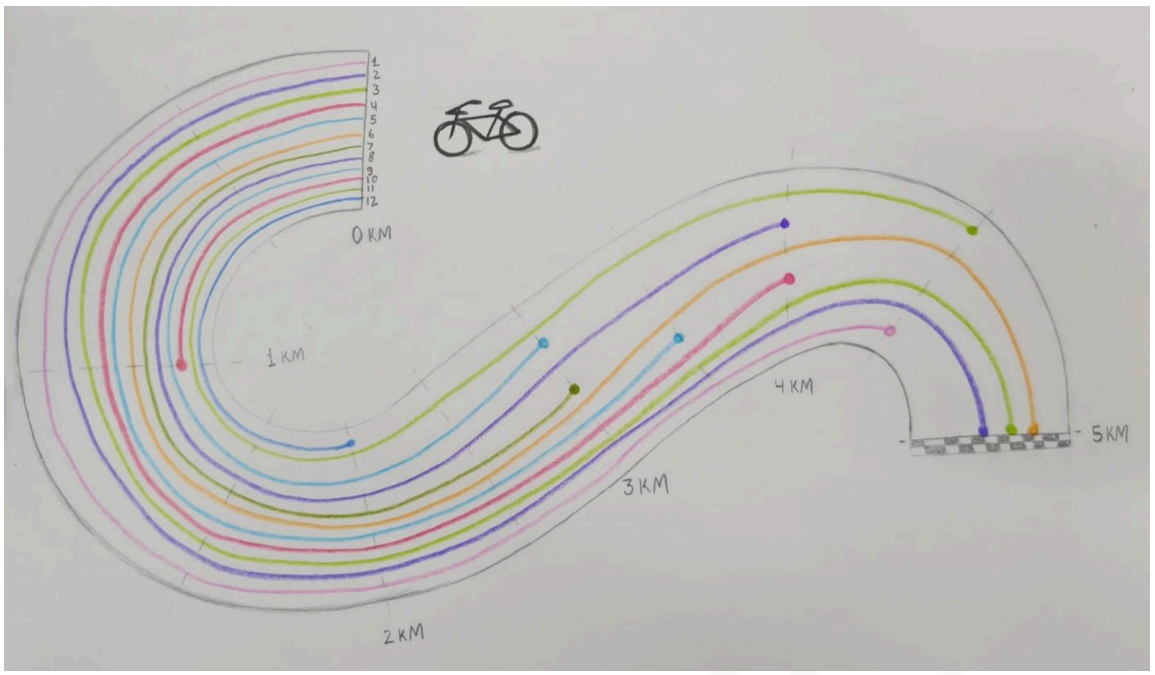
Examples of Data Visualisations

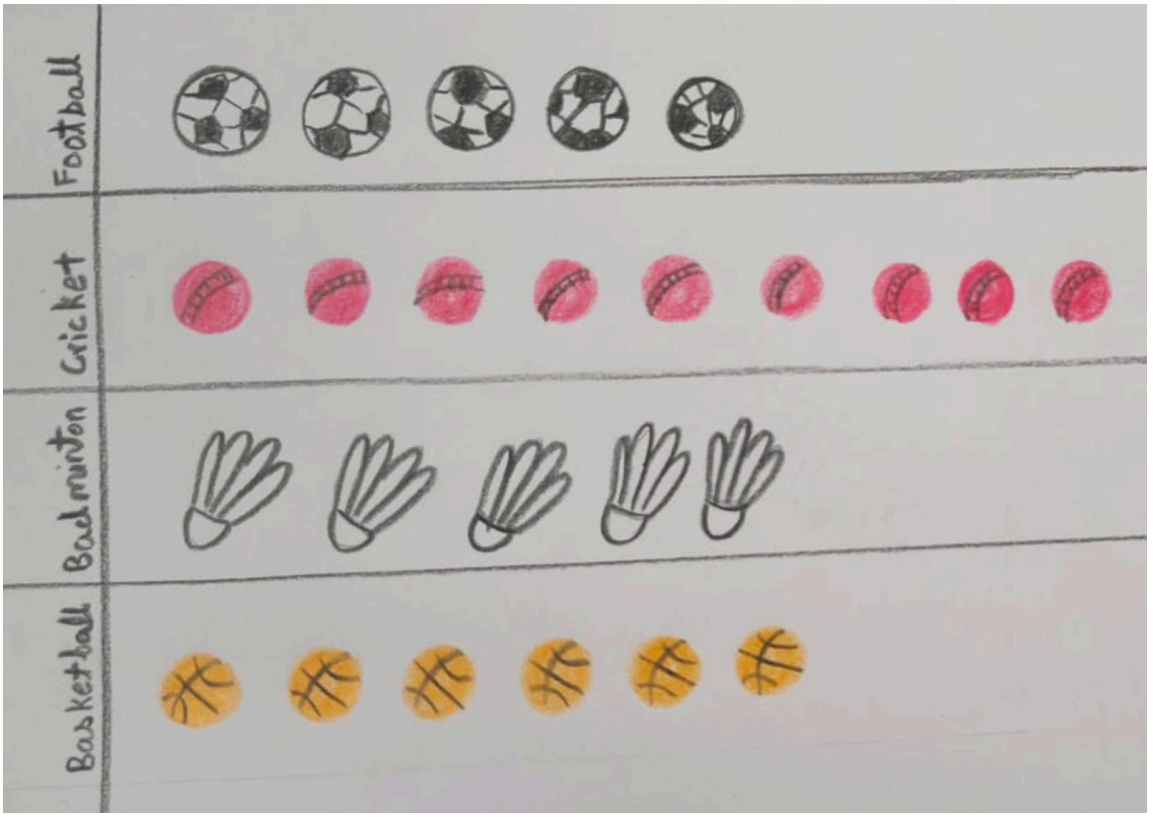
(Show it on the projector, or the teacher may take a print to display in the classroom)





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